Debt Payoff Optimization

Generated by uber-polya

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Mode: Full Pipeline

1. Problem Formulation

Problem Type: Find

Domain: Financial Mathematics

Universe

- Problem: \{debts, extra_monthly_payment, total_balance, total_minimums\}
- Avalanche: \{strategy, total_interest_paid, months_to_payoff, total_paid, is_optimal, is_feasibl
- Snowball: \{strategy, total_interest_paid, months_to_payoff, total_paid, is_optimal, is_feasible,

Variables

Symbol	Meaning	Type / Range
L	loan principal = L	\mathbb{{R}}_{>0}
r	monthly interest rate = APR/12	\mathbb{{R}}_{>0}
n	total number of monthly payments	\mathbb{{Z}}^+
PMT	monthly payment amount	\mathbb{{R}}_{>0}

Structure

You have 4 debts with different balances, interest rates, and minimum payments:

Debt	Balance	APR	Min Payment
Credit Card	$6,200	22.99%	$120/mo
Car Loan	$12,000	6.50%	$250/mo
Student Loan	$25,000	4.50%	$280/mo
Personal Loan	$3,500	15.00%	$75/mo

Total debt: $46,700. **Total minimums**: $725/mo. **Extra available**: $500/mo.

Find the optimal allocation of extra payments to minimize total interest paid over time. Compare two strategies:

1. **Avalanche** (highest interest rate first) -- mathematically optimal among greedy single-target strategies
2. **Snowball** (lowest balance first) -- psychologically motivating but costs more in interest

Real-World Mapping

Real-World Concept	Mathematical Object
loan balance	L — principal
annual percentage rate	r — monthly rate = APR/12
loan term	n — number of monthly payments
monthly mortgage payment	PMT — annuity payment

Constraints

(1)

$$PMT = L \cdot \frac{r(1+r)^n}{(1+r)^n - 1}$$

annuity payment formula

(2)

$$B_k = B_{k-1}(1+r) - PMT$$

amortization recurrence

(3) B_n = 0 *loan fully repaid at maturity*

(4)

$$\text{Total Interest} = n \cdot PMT - L$$

total interest = payments minus principal

Objective

$$\min_{\text{option}} \sum_{k=1}^{n} PMT_k \ + \ \text{closing costs}$$

Approach

Suggested approach: unknown

Complexity class:

Available tools: unknown

2. Solution

Answer:	Financial analysis complete
Optimal:	No / Unknown
Feasible:	Yes
Algorithm:	unknown
Time:	0.0000s
Certificate:	Amortization verified: final balance = $0.00 for all options.

Solution Details

Verification

No independent verification checks recorded.

3. Interpretation

The Question

You have 4 debts with different balances, interest rates, and minimum payments:

Debt	Balance	APR	Min Payment
Credit Card	$6,200	22.

The Answer

Financial analysis complete.

What This Means

- Total interest paid under each strategy
- Months to complete payoff under each strategy
- Head-to-head comparison table showing interest savings
- Payoff timeline showing when each debt reaches zero (avalanche)
- Independent verification (5 checks per strategy)

Month-by-month greedy simulation:

1. Accrue interest on all debts (monthly_rate = APR / 12)
2. Pay minimums on every debt with remaining balance
3. Allocate extra (plus freed minimums from paid-off debts) to the highest-priority target debt
4. Cascade surplus -- if the target is paid off mid-allocation, redirect remaining funds to the next priority debt
5. Repeat until all balances reach zero

The avalanche strategy prioritizes debts by descending APR, which is provably optimal because it minimizes the balance-weighted average interest rate at every step. The snowball strategy prioritizes by ascending balance for faster psychological wins.

Recommendations

1. Compare total interest paid, not just monthly payments, when evaluating mortgage options.

Limitations

- Assumes fixed interest rates; adjustable-rate mortgages introduce variability.
- Does n

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